A Policy-Aware ROS-Based Library Assistant: Deployment and Evaluation on a Pepper Robot

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***Abstract*—University libraries are high-demand service environments that combine the technical requirements of any indoor public space, such as reliable navigation and safe operation around pedestrians, with stricter norms of their own: quiet zones, predictable service workflows, and strong privacy expectations. This paper presents the design, implementation, and preliminary real-world evaluation of a proof of concept for Nova, a ROS-based library assistant deployed on a Pepper humanoid robot. The assistant resolves motion planning and navigation to areas of interest (e.g., book location by subject area, library reception desk, study rooms, restrooms, elevators, and so on), and basic service inquiries through a bilingual, tablet-centered interaction model and semi-autonomous, constrained indoor guidance, and it is governed by location-based policies that make its speech, motion, and navigation depend on the operational meaning of each area. A central design decision decouples the environment mapping from the service layer running on Pepper. Because Pepper's native sensing proved insufficient for reliable SLAM, the reference map was built at the beginning of the project using a Hiwonder MentorPi M1 platform equipped with a 360-degree LiDAR, while the service layer relies on that map to run the tasks the robot offers to the users. The Pepper robot assistant was tested in an academic library with navigation as the dominant service and a clear preference for being physically guided. The results suggest that, in rule-constrained environments, the acceptance of a social robot depends more on policy compliance and dependable navigation than on maximal autonomy.**

***Index Terms*—Social Robotics, Human-Robot Interaction, Indoor Navigation, Service Robots, ROS.**

I. Introduction

University libraries operate as high-demand service environments. Beyond providing access to physical and digital collections, they continuously support users with spatial orientation, procedural guidance, and information about academic services. Many of these requests are repetitive and time-sensitive, and although each is individually simple, together they shape the user experience: when guidance is slow or unclear, students lose time, staff workload rises, and the overall service capacity of the library is reduced. This effect is most visible during peak hours, precisely when immediate staff availability is scarcest.

Service robots have been deployed in a range of indoor public settings, such as airports, hospitals, and hotels, where they provide guidance and basic support [1], [2]. These environments share key demands with libraries, including reliable indoor navigation and safe operation around pedestrians. Libraries, however, add constraints that are decisive for system design: quiet-zone norms restrict where voice interaction is acceptable, service workflows must remain predictable for staff and users, and any data collection must respect the privacy expectations of an academic institution [3].

Against this background, this work develops a proof of concept for a Pepper-based, ROS-integrated library assistant, named Nova, that operates within these constraints. The assistant handles user inquiries, provides indoor guidance to key library locations, and supports basic service operations through a structured, tablet-centered interface. In this way, the robotic library assistant is a complement to, rather than a replacement for, the library staff. Rather than treating the library merely as a navigation problem, the design frames it as a rule-constrained public space in which technical performance must align with everyday service expectations. Unlike deployments aimed primarily at observing users' first impressions [4], the system is engineered around operational needs and prioritizes a quiet-first interaction model, so that users can request help without introducing noise into study areas. A robotic assistant that ignores these constraints may work as a demonstration, yet remain unacceptable for real operation. Furthermore, to strictly adhere to privacy standards, our system does not collect any personal data from users; all logged analytics correspond solely to the functional acceptance of the robot's tasks.

The contribution of this paper is threefold. First, it documents a practical deployment pipeline that decouples environment mapping from Pepper while preserving semi-autonomous service behavior on the target platform. Second, it defines a set of location-aware interaction policies in which speech, motion, and navigation depend on the operational meaning of each library area, and shows how these policies can be enforced by the same layer that drives the interface. Third, it reports evidence from a preliminary real-world deployment based on user-initiated sessions and usage analytics, rather than from laboratory demonstrations alone.

II. Related Work

Deploying social robots in public spaces requires addressing safety around pedestrians, robustness to changing layouts, and clear interaction boundaries [5]. Studies situated specifically in libraries are instructive: when Pepper was placed in an

academic library, students' first impressions combined novelty and curiosity with discomfort, showing that acceptance is not automatic and is strongly shaped by context and by how legible the robot's role is [4]. This motivates two design commitments adopted here, namely a structured, easily understood interaction and a modality suited to quiet spaces.

Enabling reliable autonomy on Pepper typically means integrating its native stack with established robotics middleware. Previous work has brought ROS-based localization and navigation to Pepper to achieve semi-autonomous movement among people and obstacles [6], and case studies report SLAM-based localization and navigation adapted to the platform's sensing and compute constraints [7]. Beyond point-to-point motion, operation in shared spaces calls for reactive obstacle avoidance and, in denser settings, human-aware or socially aware planning [8], [9]. Complementary efforts emphasize engineering discipline rather than one-off prototypes: structured frameworks support repeatable deployment and system-level evaluation on Pepper [10], and simulation environments enable safer, reproducible iteration before real-world trials [11], [12]. From a privacy standpoint, public-space robots should minimize data exposure and avoid passive monitoring [13], [14], which reinforces the user-initiated, tablet-centered model adopted in this work.

Although these works establish the technical feasibility of Pepper-based navigation and public-space interaction, they leave a narrower gap that is central to this paper: how to adapt a social robot to a rule-constrained academic library where assistance must remain quiet, predictable, and privacy-preserving. Our approach treats these constraints not as afterthoughts but as engineering requirements that shape both the system architecture and the evaluation criteria.

III. System Implementation

The system is organized around a tablet-hosted web interface running on Pepper, a ROS navigation layer, a waypoint-graph management tool, and a lightweight analytics backend. Fig. 1 summarizes the overall architecture. The interface is the orchestration layer: it presents service options, logs user-initiated actions, and invokes either navigation or information behaviors. The navigation layer handles localization and motion through AMCL (Adaptive Monte Carlo Localization) and the `move_base` ROS node, while a separate dashboard lets operators capture destinations and maintain the route graph. This separation of concerns is deliberate, since it allows the service logic and the navigation substrate to evolve independently.

A. Environment Mapping and Preparation

Initial laboratory tests revealed that Pepper's native laser sensing did not provide sufficiently robust coverage for high-quality SLAM; its sparse laser arrangement left gaps in obstacle detection, and an alternative depth-camera configuration proved sensitive to object height and position. Repeated trials on different testing sessions indicated that this was a structural limitation of the platform rather than an incidental execution error. Consequently, a two-robot strategy was adopted: Pepper was retained as the service platform, while mapping was reassigned to a Hiwonder MentorPi M1 equipped with a 360-degree LiDAR, which is better suited to two-dimensional environment reconstruction. The difference in sensor height between the two robots is not large enough to make the map unusable. Both the MentorPi LiDAR and Pepper's base lasers sit close to the floor, and the environment consists mostly of stable large-scale structures such as walls, corridors, and shelves, which allowed AMCL to localize reliably on this map during the real-environment tests.

The reference map was not obtained in a single pass. A diagnostic first attempt exposed the main sources of error: misalignment in the reconstruction, artifacts from transient obstacles such as moving people and chairs, and spurious returns near the glass entrance, where the sensor occasionally read beyond the physical limits of the space. From these observations a set of refinement criteria was defined, namely starting the map at the robot's intended operating point near the reception desk, following straighter trajectories to reduce distortion in long corridors, and continuously saving intermediate versions. Across a second cycle of incremental attempts, each pass corrected the errors of the previous one until one version achieved the best balance between global consistency and structural detail (Fig. 2). A further pass intended to refine the same region instead degraded areas that were already stable, so it was discarded and the previous version retained. This confirmed a practical lesson that is easy to overlook: overwriting to fix a local area can damage parts of the map that were already correct, which makes the preservation of intermediate versions essential.

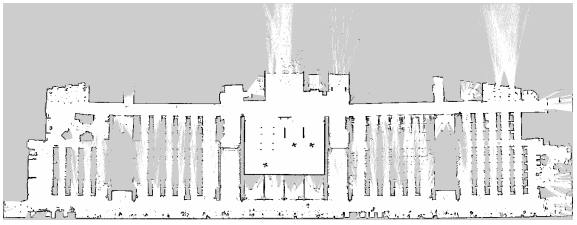


Fig. 2. The selected reference SLAM map, chosen for achieving the optimal balance between global consistency and structural detail prior to navigation cleanup.

The selected SLAM map was then specialized into several representations with distinct purposes. The original map was preserved for localization, since it retains the environmental detail AMCL needs to estimate pose. A cleaned version simplified contours and removed non-structural artifacts to support the definition of routes and points of interest. On top of this, non-relevant areas were marked as dead zones, and key destinations, such as study rooms, restrooms, the elevator, the reception desk, and the main entrance, were annotated for planning, yielding a transit map. Rather than forcing the robot to enter every physically reachable space, this map restricts movement to stable, high-value paths compatible with library policies and connectivity constraints, as shown in Fig. 3. At

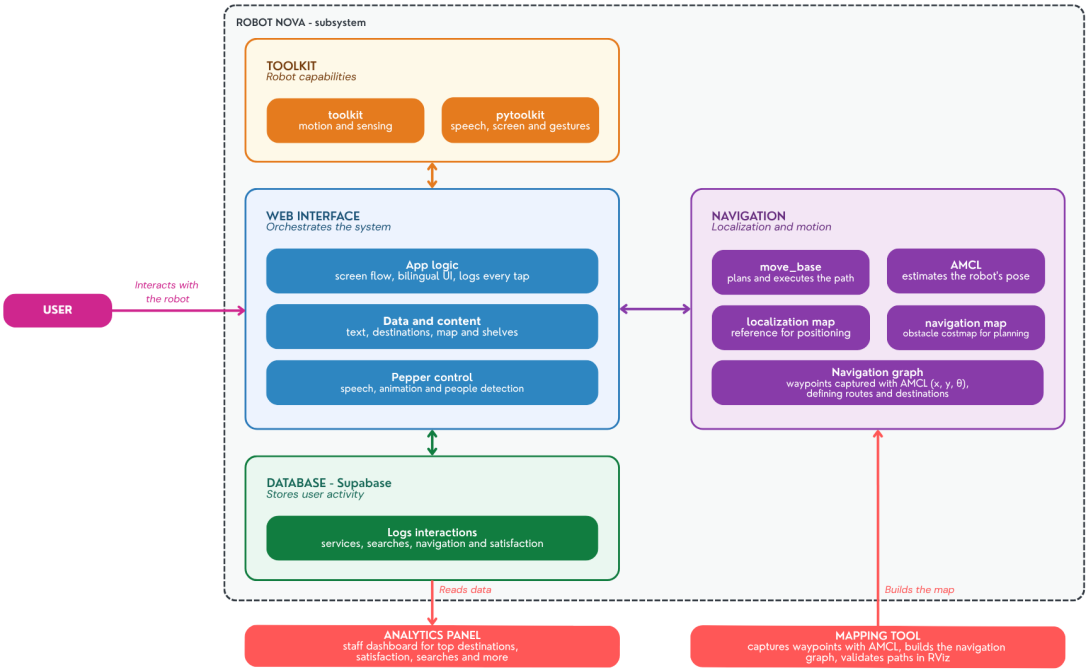


Fig. 1. System architecture integrating the tablet interface, Pepper services, ROS navigation, waypoint graph, and analytics layer.

run time, navigation actually relies on two of these maps used together: Pepper localizes on the detailed SLAM map while the cleaned navigation map, with its dead zones, is overlaid to constrain where the robot may plan and move, so that accurate localization and bounded motion are satisfied without forcing a single representation to serve both roles. Neither map is meant to capture transient obstacles. At runtime these are handled by the local costmap fed by Pepper's base lasers, whose coverage is sparse (three sensors of 15 beams each), so detection is coarse and the bumpers act as a fallback, while the dead zones keep the robot on stable corridors away from tables and chairs.

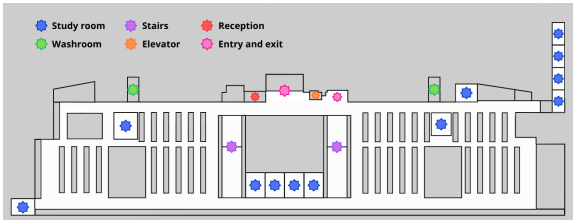


Fig. 3. Planning map derived from the SLAM output, with relevant destinations and constrained transit regions.

B. ROS-Based Navigation on Pepper

The navigation stack runs on Pepper utilizing the `move_base` node to dispatch global navigation goals and handle local reactive obstacle avoidance, working in tandem with AMCL [15]. To manage navigation goals, an internal dashboard communicates with the robot through `rosbridge`, letting operators drive the robot to a point, read its exact pose (x, y, θ) from AMCL, name it, and record how it connects to existing nodes. The outcome is a single JSON file describing the complete navigation graph, which the main application later loads. Because a destination's real coordinates depend on how the map was generated and on where the base physically stands, this capture-and-validate procedure is more reliable than entering coordinates by hand, and it makes the map easy to maintain: adding or adjusting a destination only requires updating the graph rather than redefining the application logic.

Reliable motion in the real environment required several corrective measures that are often omitted in high-level architecture descriptions but proved decisive in practice. Movement between nodes is generated by the path planner to ensure safety and reliability. The robot autonomously tracks the planned route, adjusting its heading at key intersections and leveraging localized orientation references at each node to maintain precise alignment. Two further issues were addressed: the robot's arms swung on their own during transit, which was solved by explicitly disabling arm motion while navigating and restoring the idle animation on arrival; and the base introduces a small cumulative offset, which is currently mitigated by setting the initial pose manually in RViz at the start of operation. Finally, two background routines keep the robot stable, periodically clearing the local obstacle costmap to remove spurious readings and restoring the standing posture between interactions.

C. Web Interface and Interaction Flow

The user interface, hosted on Pepper's tablet, orchestrates the system through a bilingual (Spanish/English) frontend that translates user taps into navigation commands; the language can be switched at any point during a session. The interaction is policy-driven: the robot produces spoken output only when a

request is resolved, for instance when it begins guiding a user or reaches a destination, and remains silent while navigating or waiting, honoring the library's quiet zones. From the visitor's perspective, the session begins with language selection and a service menu that groups navigation, book topic search, book services, general information, and events. For a navigation request, the interface shows the destination on the library map and offers two options, physical guidance by the robot (Fig. 4) or path visualization on the tablet alone, a distinction that proved useful because some users preferred orientation without walking alongside the robot.

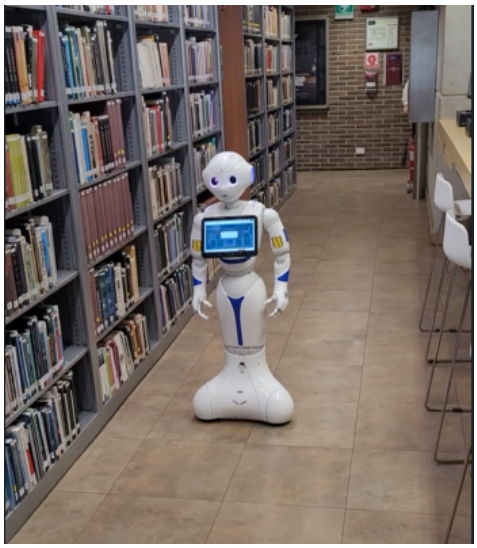


Fig. 4. Pepper during real-environment guided navigation tests in the library.

Crucially, the same layer that presents these options also enforces the environment's rules, so the interface acts simultaneously as a service frontend and a policy-enforcement point. Table I summarizes the location-based policies. A session always starts and ends at a specific point near the reception desk, keeping the robot's behavior predictable for staff. Nearby destinations that are visible from the base are resolved without displacement: the robot points toward the place with a spoken indication and a gesture. Spoken outputs rely on hard-coded text-to-speech templates that dynamically respond to the robot's state, while arm kinematics for pointing utilize pre-calculated joint angles mapped to the target's relative direction. Destinations near weak-connectivity areas are approached only up to the closest reliable node, after which Pepper triggers a specific explanatory speech template detailing the remaining path to the user, and still asks the user for a rating, so a connectivity limit is handled as a first-class case rather than a failure. Finally, a help button remains available throughout the interface for situations the robot cannot resolve on its own; pressing it sends an email to library staff, phrased as if the robot itself was requesting assistance, and accompanied by the visitor's most recent request. The robotic assistant therefore always preserves a call back to a human, consistent with its role of supporting library staff rather than standing in for them.

TABLE I
LOCATION-BASED INTERACTION POLICIES

Zone / case	**Spoken output**	**Planning and Navigation**
Reception desk (base)	Opens and closes each session	Start and return point of every interaction
Study zones	Silent during transit	Navigate through permitted corridors
Destination visible from base	Indicates direction by voice and gesture	No displacement
Weak-connectivity area	Explains the remaining path via template	Stop at the closest reliable node

D. Deployment Workflow

Once the maps and the navigation graph are in place, operation follows a consistent workflow. After powering the robot and starting its service toolkit, the localization and navigation stack is launched and the robot's pose is initialized in RViz. From that point the tablet interface becomes the main entry point for users. Every completed task ends with a rating request, after which the robot returns to its base at the reception desk, leaving the system in a clean state for the next visitor.

IV. RESULTS AND DISCUSSION

The system was tested with real users in the university library, in what was the first end-to-end use of the assistant by people outside the development team. Data was collected exclusively from user-initiated interactions and aggregated into a usage analytics dashboard (Fig. 5).

A. Usage and Service Demand

The deployment took place over four operating days between June 10 and June 15, 2026. The assistant was available only during staffed sessions of a few hours each, so session counts reflect the robot's availability rather than library-wide demand. During this period, the system registered 73 user sessions, where a session denotes a user-initiated task rather than a distinct person, so a single visitor performing several actions produces several sessions. Wayfinding was the most-demanded service: the *Where do I go?* option was opened in 42 of the 73 sessions (58%), followed by book topic search with 10 sessions, interactive events with 9, information queries with 7, and book services with 5; Table II reports the corresponding shares. Both wayfinding and book search lead to a physical location, so 52 sessions (71%) could yield a destination, and 40 (55%) selected one. Of those 40, guidance was preferred: 28 (70%) were led by the robot and 12 (30%) consulted the map alone. The most frequently requested destinations were the reception desk (return point of each user session), the women's restroom, and study room 255D. Among information queries, engineering drawing and artificial intelligence were the most searched topics. The analytics backend logs a session only when a task reaches the rating step. Aborted interactions,

help-button presses, and connectivity-limited stops were not instrumented in this deployment, so the 73 sessions represent completed tasks and no reliability counts can be derived from the logs.

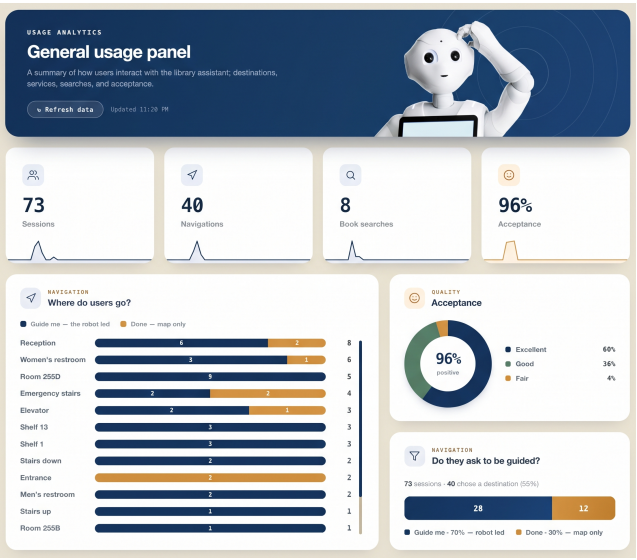


Fig. 5. Administrative dashboard summarizing sessions, destinations, and acceptance outcomes from the deployment in a library. Only rating levels with responses are displayed, and the book search counter (8) reports completed searches rather than sessions that opened the option (10).

These distributions carry two implications. First, the dominant demand in this context is spatial assistance rather than conversational novelty, which validates the focus on navigation as the core service. Second, a tablet-first interaction does not suppress engagement; instead, it offers a low-noise way to request help while still allowing the robot to deliver a socially visible guidance service when the user wants it. Because the analytics count user-initiated tasks rather than persons, these figures describe service demand in a form the library can use for staffing and planning, and they do so without any passive monitoring, consistent with the privacy objective of the design.

B. User Acceptance and Reliability

Upon completing a guidance task or answering a query, the robot asked users to rate their perception of how good the robot accomplished the task or query through the tablet's five-level scale (excellent, good, fair, bad, and very bad). Because it is a self-report solicited by the robot immediately after a completed task, it may lean toward positive responses, so it is read as an indicator of acceptance rather than as a measure of service quality. Ratings concentrated on the upper end of the scale: 60% excellent, 36% good, and 4% fair, with no responses in the two lowest levels, for an overall 96% positive rating in the top two levels (Table II). The free-text comments were short and positive, typically brief thanks after arriving at a destination. Qualitatively, the corrective measures described above, namely the explicit control of Pepper's arms, the tuning of the arrival orientation at each node, and the continuous clearing of the obstacle costmap, produced navigation that was noticeably more stable and repeatable, mitigating the localization drift and trajectory instability observed in earlier runs. Beyond the metrics, the deployment also met a softer goal that is easy to see but hard to quantify: visitors who were hesitant at first grew comfortable with the robot over the course of a session, which is itself relevant to acceptance in everyday spaces.

TABLE II
SUMMARY OF USAGE METRICS

Metric	Value
Total User Sessions (Completed Tasks)	**73**
Service Breakdown	**Sessions (%)***
Where do I go?	42 (58%)
Book Topic Search	10 (14%)
Interactive Events	9 (12%)
General Information Queries	7 (10%)
Book Services	5 (7%)
Destination Requests	**Sessions (%)**
Sessions Selecting a Destination	**40 (55%)**
– Guided by the robot	28 (70% of 40)
– Map only	12 (30% of 40)
User Acceptance Indicator	**Sessions (%)**
Overall Rating	**70 (96% positive)**
– Excellent	44 (60%)
– Good	26 (36%)
– Fair	3 (4%)
– Bad	0 (0%)
– Very bad	0 (0%)

*Percentages rounded; session counts sum to 73.

C. Scope of the Evaluation

The evidence reported here characterizes adoption and functional acceptance rather than raw navigational performance. Formal metrics, such as task-completion rate, traversal time, and localization-failure counts, were not quantified under controlled trials, and standardized instruments such as SUS or the Godspeed questionnaires were not applied. We therefore frame this work as a deployment proof of concept study whose contribution is a working, policy-aware system and its usage evidence in a real setting. A controlled navigation evaluation is the natural next step and is discussed in Section V.

D. Discussion

Taken together, the results suggest that social robots in academic service environments should be designed around role clarity rather than maximal autonomy. Pepper was effective here because it handled a bounded set of repetitive tasks, deferred exceptional situations to staff, and exposed its capabilities through a legible tablet interface. From an engineering perspective, the most consequential lesson is that acceptance depended less on adding expressive behaviors and more on eliminating small operational failures: stable arrival poses, quiet behavior during transit, controlled arm posture, and predictable recovery from transient obstacles appeared to contribute to perceived quality. Disabling arm motion during

transit was a deliberate choice that prioritizes stable, predictable movement and safety around pedestrians, at the cost of some expressiveness, which resumes upon arrival. Policy compliance and navigation reliability were thus not separate concerns but two sides of the same user experience, and this supports a practical view of human-robot collaboration in libraries in which the robot reduces repetitive wayfinding load while institutional rules continue to define when and how it acts.

V. Conclusions and Future Work

This work suggests that adapting a social robot to a library environment is, above all, a matter of prioritizing operational policies over raw technical capability. By decoupling the mapping hardware from the service platform, defining location-based interaction policies, and centering the interaction on a quiet, tablet-first interface, the assistant reached a 96% positive acceptance rate among users for completed tasks in a real academic library while respecting the norms of the space. The deployment is consistent with the design hypothesis that, in a rule-constrained environment, the acceptance of a social robot rests on policy compliance and dependable navigation rather than on autonomy or novelty for their own sake. Beyond a single prototype, the work leaves an integrated and reproducible baseline, combining mapping, waypoint capture, a policy-driven interface, and privacy-respecting analytics, from which the assistant can be sustained and extended.

Several directions would strengthen the work. Because the current deployment lacks formal field testing methodologies and represents a preliminary proof of concept, a major next step is conducting controlled evaluations that report task-completion rates, traversal times, and failure and recovery counts. This should be matched with formal baseline comparisons, evaluating the robotic assistant against static tablet kiosks or simple mobile apps to measure the physical robot's added value, as well as benchmarking against a standard, non-policy-aware `move_base` stack. It should also be complemented by standardized usability and perception instruments such as SUS, RoSAS, and Godspeed, so that the deployment evidence presented here is paired with objective performance measures.

Regarding the system, there are several potential improvements. Because the robot is currently not fully autonomous regarding initial location recovery, an automated base-calibration process must be implemented to correct the AMCL offset. One possibility is to explore a relocalization mechanism based on visual landmarks placed near the robot's initial location so it can independently reset itself upon completing a task. Second, offline caching of static assets and locally served data would increase the resilience of the tablet interface to network instability and reduce the connectivity-limited stops. Third, extending the assistant to the remaining floors of the library would test the pipeline at a larger scale.

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